

The Allie Scenario Prompting Protocol: A Scenario-Constrained, Outcome-First Method for Prompt Engineering

Preprint manuscript prepared for Zenodo upload

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Version: 1.0 **Date:** 2026

Document type: Preprint

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Suggested citation

Allie, Darren L. *The Allie Scenario Prompting Protocol: A Scenario-Constrained, Outcome-First Method for Prompt Engineering*. Azure & Verdant Vistas, LLC, 2026. Preprint.

Attribution statement

The Allie Scenario Prompting Protocol (ASPP) and the S.C.E.N.A.R.I.O. framework are attributed to Darren L. Allie, Director of AI Workflow Optimization and AI Business Improvement, Azure & Verdant Vistas, LLC.

DOI: 10.5281/zenodo.20596641

Abstract

Prompt engineering has become an important practice for improving the usefulness, reliability, and task alignment of large language model outputs. Existing prompting methods have advanced reasoning, planning, role alignment, tool use, output structuring, and multi-path problem solving. However, many scenario-based user prompts still produce outputs that are too broad, too exploratory, or too option-heavy when the user needs a clear and actionable decision. This preprint introduces The Allie Scenario Prompting Protocol (ASPP), a scenario-constrained, outcome-first prompting methodology created by Darren L. Allie, Director of AI Workflow Optimization and AI Business Improvement, Azure & Verdant Vistas, LLC. ASPP does not claim to invent the general phrase “scenario prompting” or the broad practice of using scenarios in prompts. Instead, it formalizes a distinct approach to scenario-based prompting that emphasizes desired-outcome identification, context clarification, constraint and risk analysis, decision-space narrowing, logical reasoning, optional coding or calculation, single-best-output recommendation, conditional alternative disclosure, and executable action planning. ASPP is organized through the acronym S.C.E.N.A.R.I.O.: State the Desired Outcome, Clarify the Scenario Context, Examine Constraints and Risks, Narrow the Decision Space, Analyze with Logic, Reasoning, Coding, or Calculation, Recommend the Best Single Output, Indicate Additional Outcomes Only When Necessary, and Output an Executable Action Plan. Its central contribution is decision constriction: reducing brainstorming noise in situations where the user needs one best actionable recommendation rather than many possible options.

Keywords: prompt engineering; scenario prompting; scenario-based prompting; large language models; LLMs; AI workflow optimization; decision support; structured reasoning; decision narrowing; ASPP; S.C.E.N.A.R.I.O.

Repository / DOI: To be assigned after publication or archive deposit.

1. Introduction

Large language models are increasingly used for professional decision support, business analysis, project planning, operational troubleshooting, financial reasoning, technical drafting, and workflow design. In these contexts, users often provide scenario-type prompts: descriptions of real-world situations involving stakeholders, constraints, urgency, risk, and desired outcomes. These prompts commonly ask the model what should be done, what the best course of action is, how to respond to a problem, or how to choose among competing options.

Although modern language models can generate detailed responses, breadth can become a weakness when a user needs a decision. A scenario prompt may receive a long list of possible actions, broad brainstorming, or alternatives that are not clearly ranked. This can increase cognitive load and delay execution. This paper calls that effect brainstorming noise: output volume that appears helpful but does not efficiently direct the user toward a best actionable path.

ASPP is introduced to address this problem. It is designed for scenario prompts where the user needs a practical recommendation, not merely more possibilities. It formalizes a scenario-based prompting method that constrains the AI system to identify the desired outcome, clarify the situation, examine constraints and risks, narrow the decision space, analyze the best path, and produce one strongest actionable output unless alternatives are materially necessary.

ASPP is a bounded contribution to prompt engineering. It does not claim that the phrase scenario prompting is new, nor does it claim that scenario-based prompting has never been used. The contribution is the formalization of scenario-based prompting into an outcome-first decision protocol centered on one-best-output discipline and reduction of brainstorming noise.

2. Related Work and Literature Positioning

ASPP is positioned within a broader prompt-engineering literature that includes reasoning, planning, tool-use, and structured prompting methods. Chain-of-Thought prompting encourages intermediate reasoning steps to improve complex reasoning tasks. Plan-and-Solve prompting decomposes tasks into a plan and then executes subtasks. ReAct combines reasoning traces with actions that allow a model to interact with external sources or environments. Tree of Thoughts expands problem solving by allowing exploration of multiple reasoning paths. Prompt-pattern catalogs and surveys document reusable prompting structures and taxonomies.

ASPP differs from these methods by focusing on final-output decision constriction rather than reasoning expansion. It may use reasoning, planning, comparison, calculation, or tool use when useful, but its distinctive requirement is to narrow the final user-facing response to one best actionable recommendation unless materially justified alternatives must be disclosed.

3. Formal Definition of ASPP

The Allie Scenario Prompting Protocol (ASPP) is a scenario-based prompt-engineering protocol that formalizes how AI systems should interpret, analyze, and respond to real-world or hypothetical scenarios by identifying the desired outcome, clarifying context, examining constraints and risks, narrowing the decision space, applying reasoning or calculation where needed, and producing one best actionable output to reduce brainstorming noise.

The protocol is intended for decision-support prompts in which the user needs a recommendation, sequence, plan, or prioritized action rather than an unconstrained list of possibilities. It is not intended to replace open brainstorming, creative ideation, or exploratory research prompts where multiple divergent options are the desired output.

4. The S.C.E.N.A.R.I.O. Framework

ASPP operates through eight required stages:

Stage	Meaning	Function
S	State the Desired Outcome	Identify the result the user wants to achieve.
C	Clarify the Scenario Context	Summarize relevant facts, stakeholders, timing, and assumptions.
E	Examine Constraints and Risks	Identify limits, failure points, uncertainty, and material risks.
N	Narrow the Decision Space	Remove weak, unrealistic, redundant, or low-value options.
A	Analyze with Logic, Reasoning, Coding, or Calculation	Apply the appropriate level of structured analysis.
R	Recommend the Best Single Output	Provide one strongest actionable recommendation by default.
I	Indicate Additional Outcomes Only When Necessary	Disclose alternatives only when they are materially justified.
O	Output an Executable Action Plan	Translate the recommendation into practical next steps.

The S.C.E.N.A.R.I.O. sequence is designed to prevent the model from over-generating options and to help the user move from situation description to executable decision.

5. Decision-Constriction Model

ASPP's central model is decision constriction. Decision constriction means intentionally reducing the final response space from many possible outputs to one best actionable recommendation unless alternatives are necessary. The model can be summarized as:

Scenario Input -> Desired Outcome -> Context Clarification -> Constraint and Risk Analysis -> Option Elimination -> Reasoned Evaluation -> Best Single Recommendation -> Executable Action Plan

This model distinguishes ASPP from exploratory prompting methods. Exploratory prompting seeks breadth. ASPP seeks decision clarity. In practice, ASPP is most useful where a user faces competing priorities, financial tradeoffs, operational risk, project constraints, compliance exposure, or time-sensitive decisions.

6. Output Rules

ASPP includes three output rules:

- One-Best-Output Rule:** return one best actionable recommendation by default.
- Conditional Alternative Rule:** disclose alternatives only when they are materially close, required for safety or uncertainty, or explicitly requested.
- Executable Action Rule:** convert the recommendation into a practical plan with next steps and review criteria where relevant.

These rules are designed to reduce brainstorming noise while preserving responsible uncertainty disclosure.

7. ASPP-Bench and Evaluation Method

ASPP is paired with a benchmark-style evaluation package called ASPP-Bench. The source benchmark document, ASPP Official Benchmark Evaluation Suite v1.1, defines ASPP-SCENARIO-001: Structured Decision Optimization Evaluation. It states that the purpose is to evaluate whether an AI system correctly implements ASPP through complete execution of the S.C.E.N.A.R.I.O. framework. It also adds evaluator scoring instructions, calibration example types, and an explicit anti-fabrication rule.

The initial benchmark case asks a model to select one of three startup growth initiatives given a \$50,000 budget, 12 months of runway, limited engineering resources, above-average customer satisfaction, and no dedicated sales team. The model must apply the full S.C.E.N.A.R.I.O. framework and recommend a single best course of action.

Component	Points
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State Outcome	10
Clarify Context	10
Constraints and Risks	15
Narrow Decision Space	15
Analysis	20
Recommendation	10
Additional Outcomes	5
Action Plan	15
Total	100

ASPP-Bench classifies scores of 90-100 as exceptional compliance, 80-89 as strong compliance, 70-79 as moderate compliance, 60-69 as weak compliance, and below 60 as non-compliant. Automatic failure conditions include missing any S.C.E.N.A.R.I.O. stage, providing multiple primary recommendations, providing no reasoning, making unsupported claims, providing no action plan, ignoring scenario constraints, or fabricating facts, metrics, market data, financial projections, or assumptions not provided in the scenario.

8. Use Cases

ASPP is most appropriate for scenario prompts that require decision support rather than open-ended ideation. Suitable domains include:

- Business strategy and operational planning.
- Financial sequencing, credit, and lending decisions.
- Project-management delivery under constraints.
- Labor, staffing, and workflow optimization.
- Real estate and investment tradeoff decisions.
- Compliance-sensitive scenario analysis where uncertainty must be acknowledged.
- AI workflow design and enterprise decision-support templates.

ASPP should not be the primary method when the user asks for broad brainstorming, creative exploration, fiction writing, visual variation, or a long list of unranked possibilities. In those cases, ASPP may be used later to narrow options into one recommendation.

9. Applied Example: Startup Growth Initiative

A startup has a \$50,000 budget and must choose between hiring a full-time salesperson, increasing paid advertising spend, or developing a customer referral program. It has 12 months of runway, limited engineering resources, strong customer satisfaction, and no dedicated sales team.

Under ASPP, the desired outcome is to select the growth initiative that maximizes growth while minimizing execution risk and preserving capital. The context includes the budget, runway, customer satisfaction, engineering constraint, and lack of sales capacity. Key risks include budget exhaustion from hiring, inefficient ad spend, development burden, and limited experimentation time. Narrowing the decision space suggests delaying the full-time sales hire and treating paid advertising as a secondary test rather than the first major investment. The best primary recommendation is to develop a customer referral program, provided it can be launched with lightweight tooling and measured with clear KPIs. The action plan should include defining the referral offer, launching to satisfied customers, tracking referrals and conversion, reviewing results within a defined period, and scaling only if the program shows traction.

10. Limitations

ASPP has limitations. First, it depends on the quality of the scenario information provided. Missing facts may require assumptions, and assumptions must be labeled. Second, the one-best-output rule may be inappropriate when the user's goal is creativity, exploration, research discovery, or brainstorming. Third, ASPP should not be treated as a substitute for professional judgment in financial, legal, medical, safety, employment, or regulated contexts. Fourth, ASPP-Bench has been defined as a benchmark-style evaluation suite, but wider empirical validation across multiple models and evaluators remains future work.

11. Future Work

Future work should focus on publishing this preprint, archiving the supporting ASPP materials, expanding ASPP-Bench beyond the initial benchmark case, creating full calibration examples, testing ASPP across multiple LLM systems, releasing prompt templates, creating a public repository, and collecting community feedback. A later ASPP v1.1 should incorporate benchmark results, inter-rater reliability procedures, additional scenario categories, and platform-specific prompt templates.

12. Conclusion

The Allie Scenario Prompting Protocol (ASPP) is introduced as a scenario-constrained, outcome-first method for prompt engineering. It formalizes scenario-based prompting into a structured decision-support protocol that clarifies the desired outcome, examines context and constraints, narrows the decision space, applies reasoning or calculation where needed, recommends one best actionable output, discloses alternatives only when necessary, and ends with an executable action plan. ASPP's central contribution is decision constriction: reducing brainstorming noise when the user needs a clear recommendation rather than a broad set of possibilities. By pairing the protocol with ASPP-Bench and a scoring rubric, ASPP is positioned as a testable and reusable proposed formalization of scenario-based prompting.

13. Zenodo Deposit Metadata

Field	Recommended Entry
Resource type	Publication - Preprint
Title	The Allie Scenario Prompting Protocol: A Scenario-Constrained, Outcome-First Method for Prompt Engineering
Creator	Allie, Darren L.
Affiliation	Azure & Verdant Vistas, LLC
Version	1.0
Language	English
License	Creative Commons Attribution 4.0 International (CC BY 4.0)
Keywords	prompt engineering; scenario prompting; scenario-based prompting; large language models; AI workflow optimization; decision support
Related identifiers	DOI or repository link to be added after deposit, if available

This metadata section is included to support accurate Zenodo upload and citation. A DOI should be added to the record and future PDF version after Zenodo assigns or reserves it.

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